Business Problem:

Jamboree, a leading educational institution, has a strong track record of helping students secure admissions to top global universities through effective test preparation strategies for GMAT, GRE, and SAT. To further enhance student support, Jamboree has introduced an admission probability assessment feature on its website, specifically tailored for Indian applicants aiming for Ivy League colleges.

1.EDA

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

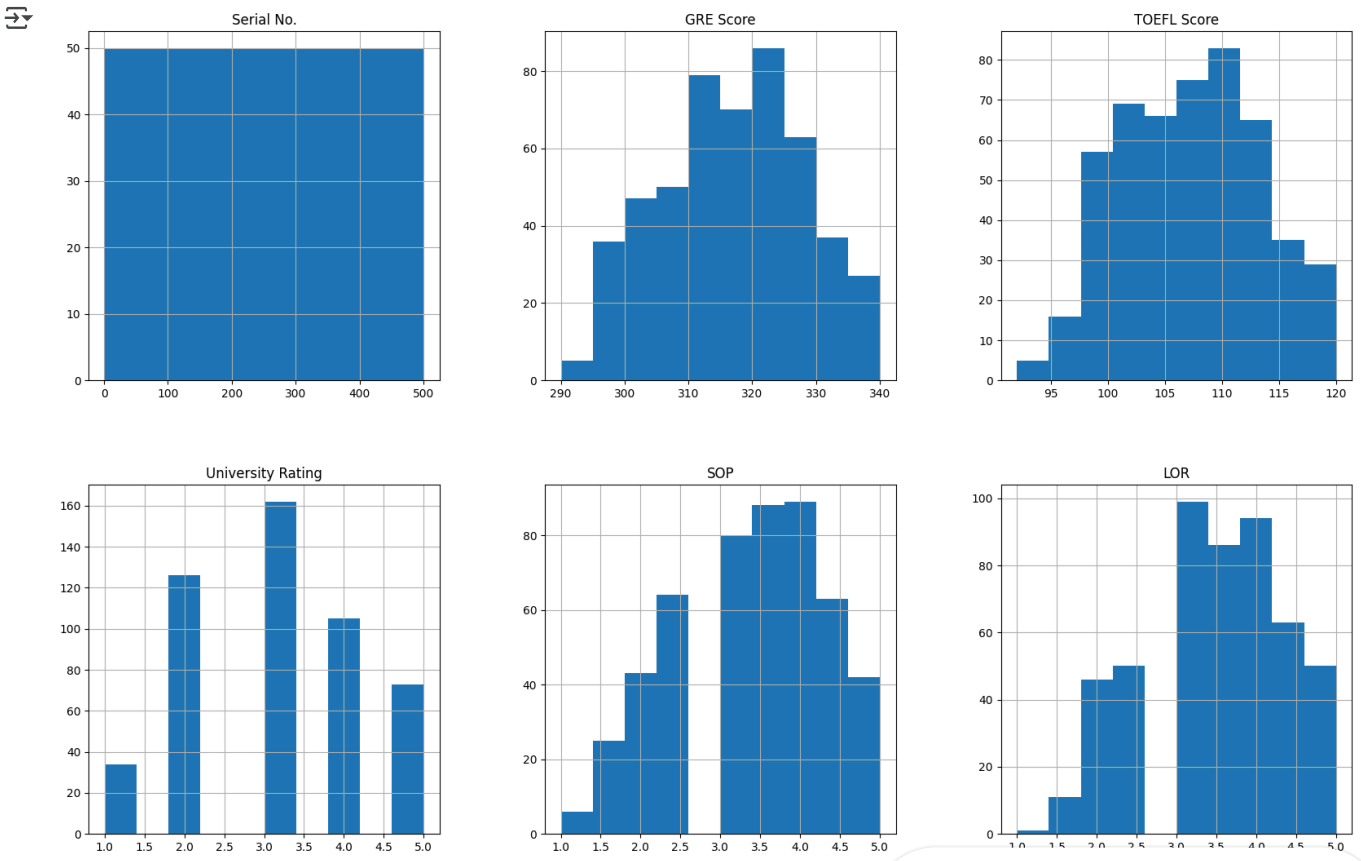
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

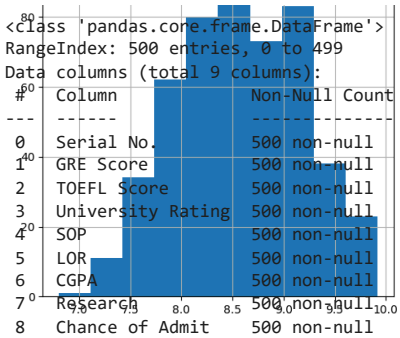
```
df = pd.read_csv("/content/drive/MyDrive/Sriram_datasets/Jamboree_Admission.csv")
df.head()
```

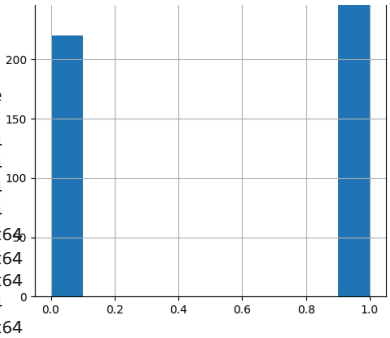
	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
0	1	337	118	4	4.5	4.5	9.65	1	0.92
1	2	324	107	4	4.0	4.5	8.87	1	0.76
2	3	316	104	3	3.0	3.5	8.00	1	0.72
3	4	322	110	3	3.5	2.5	8.67	1	0.80
4	5	314	103	2	2.0	3.0	8.21	0	0.65

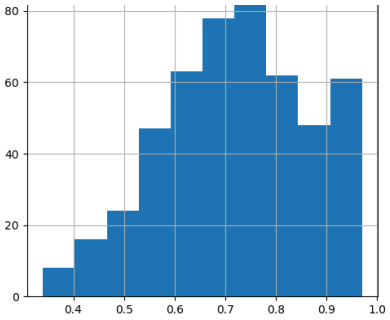
```
#df = df.drop_duplicates(subset='Serial No.')
df.hist(figsize = (20,20))
plt.show()
```




```
print(df.info())
```







```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Serial No.         500 non-null    int64  
 1   GRE Score          500 non-null    int64  
 2   TOEFL Score        500 non-null    int64  
 3   University Rating  500 non-null    int64  
 4   SOP                500 non-null    float64
 5   LOR                500 non-null    float64
 6   CGPA               500 non-null    float64
 7   Research           500 non-null    int64  
 8   Chance of Admit    500 non-null    float64
dtypes: float64(4), int64(5)
memory usage: 35.3 KB
None
```

```
df.describe(include = 'all')
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.00000	500.000000	500.000000	500.00000
mean	250.500000	316.472000	107.192000	3.114000	3.374000	3.48400	8.576440	0.560000	0.72174
std	144.481833	11.295148	6.081868	1.143512	0.991004	0.92545	0.604813	0.496884	0.14114
min	1.000000	290.000000	92.000000	1.000000	1.000000	1.00000	6.800000	0.000000	0.34000
25%	125.750000	308.000000	103.000000	2.000000	2.500000	3.00000	8.127500	0.000000	0.63000
50%	250.500000	317.000000	107.000000	3.000000	3.500000	3.50000	8.560000	1.000000	0.72000
75%	375.250000	325.000000	112.000000	4.000000	4.000000	4.00000	9.040000	1.000000	0.82000
max	500.000000	340.000000	120.000000	5.000000	5.000000	5.00000	9.920000	1.000000	0.97000

observations:

1. out of 500 students ,the average gre score is 316
2. out of 500 students ,the average toe score is 107
3. students whose avg cgpa is 8.5 ,their avg gre and toe scores are in the range of 316 and 107 with 72% chance of admit with university rating 3.1

```
print(df.isnull().sum())
```

```
Serial No.           0
GRE Score            0
TOEFL Score          0
University Rating    0
SOP                  0
LOR                  0
CGPA                 0
Research             0
Chance of Admit      0
dtype: int64
```

There are no null null values in df.

```
df.columns
```

```
Index(['Serial No.', 'GRE Score', 'TOEFL Score', 'University Rating', 'SOP',
       'LOR', 'CGPA', 'Research', 'Chance of Admit'],
      dtype='object')
```

```
df.value_counts()
```

Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit	count
1	337	118	4	4.5	4.5	9.65	1	0.92	1
330	297	96	2	2.5	1.5	7.89	0	0.43	1
343	308	106	3	3.0	3.0	8.24	0	0.58	1
342	326	110	3	3.5	3.5	8.76	1	0.79	1
341	312	107	3	3.0	3.0	8.46	1	0.75	1
...	...	...	...	...	...	...	...	...	...
162	298	99	1	1.5	3.0	7.46	0	0.53	1
161	315	103	1	1.5	2.0	7.86	0	0.57	1
160	297	100	1	1.5	2.0	7.90	0	0.52	1
159	306	106	2	2.0	2.5	8.14	0	0.61	1
500	327	113	4	4.5	4.5	9.04	0	0.84	1

500 rows × 1 columns

```
df['SOP'].value_counts()
```

SOP	count
4.0	89
3.5	88
3.0	80
2.5	64
4.5	63
2.0	43
5.0	42
1.5	25
1.0	6

```
numerical_features = [feature for feature in df.columns if df[feature].dtypes!='o']
print('Number of numerical variables: ', len(numerical_features)) # This line was incorrectly indented
# visualise the numerical variables
df[numerical_features].head()
```

```
Number of numerical variables:  9
```

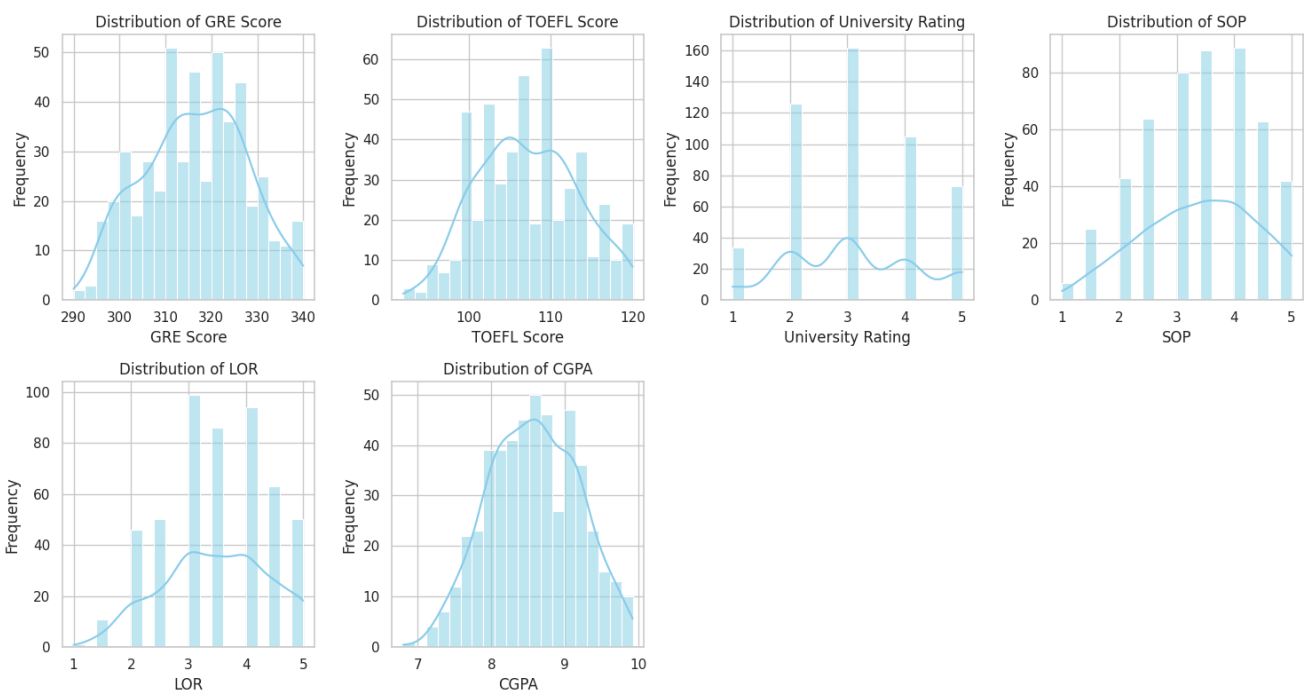
	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
0	1	337	118	4	4.5	4.5	9.65	1	0.92
1	2	324	107	4	4.0	4.5	8.87	1	0.76
2	3	316	104	3	3.0	3.5	8.00	1	0.72
3	4	322	110	3	3.5	2.5	8.67	1	0.80
4	5	314	103	2	2.0	3.0	8.21	0	0.65

In this dataset all are numerical features

Univariate Analysis

```
# Import necessary libraries for plotting
import matplotlib.pyplot as plt
import seaborn as sns
# Set the style for better visualization
sns.set(style="whitegrid")
```

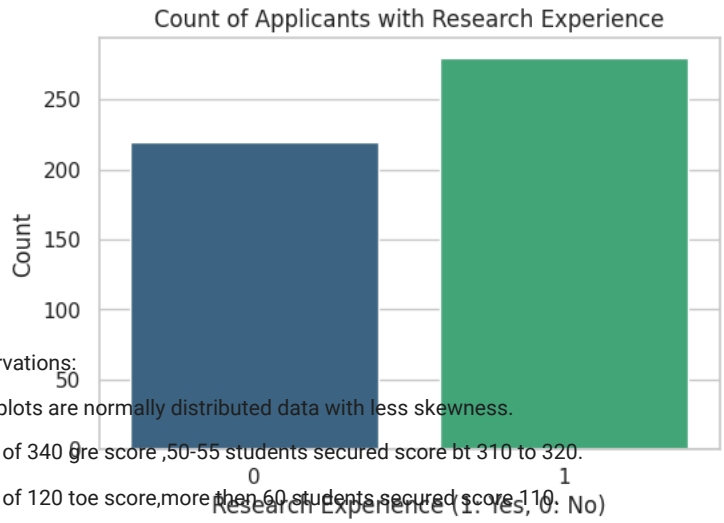
```
# Define continuous and categorical variables
# Removed the extra space after 'LOR'
continuous_vars = ['GRE Score', 'TOEFL Score', 'University Rating', 'SOP', 'LOR', 'CGPA']
categorical_vars = ['Research']
# Univariate analysis - Distribution plots for continuous variables
plt.figure(figsize=(15, 8))
for i, var in enumerate(continuous_vars, 1):  # for index, value in enumerate(fruits, start=1)
    plt.subplot(2, 4, i)
    sns.histplot(df[var], kde=True, bins=20, color='skyblue')
    plt.title(f'Distribution of {var}')
    plt.xlabel(var)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
# Univariate analysis - Barplot for categorical variables
plt.figure(figsize=(6, 4))
sns.countplot(x='Research', data=df, palette='viridis')
plt.title('Count of Applicants with Research Experience')
plt.xlabel('Research Experience (1: Yes, 0: No)')
plt.ylabel('Count')
plt.show()
```




```
<ipython-input-19-ffe20013249d>:22: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(x='Research', data=df, palette='viridis')
```



observations:

1.All plots are normally distributed data with less skewness.

2.out of 340 gre score ,50-55 students secured score bt 310 to 320.

3.out of 120 toe score,more then 60 students secured score 110.

4.Around 160 students got admission with university rating 3.

5.>80 students whose sop in range 3.5 to 4 .

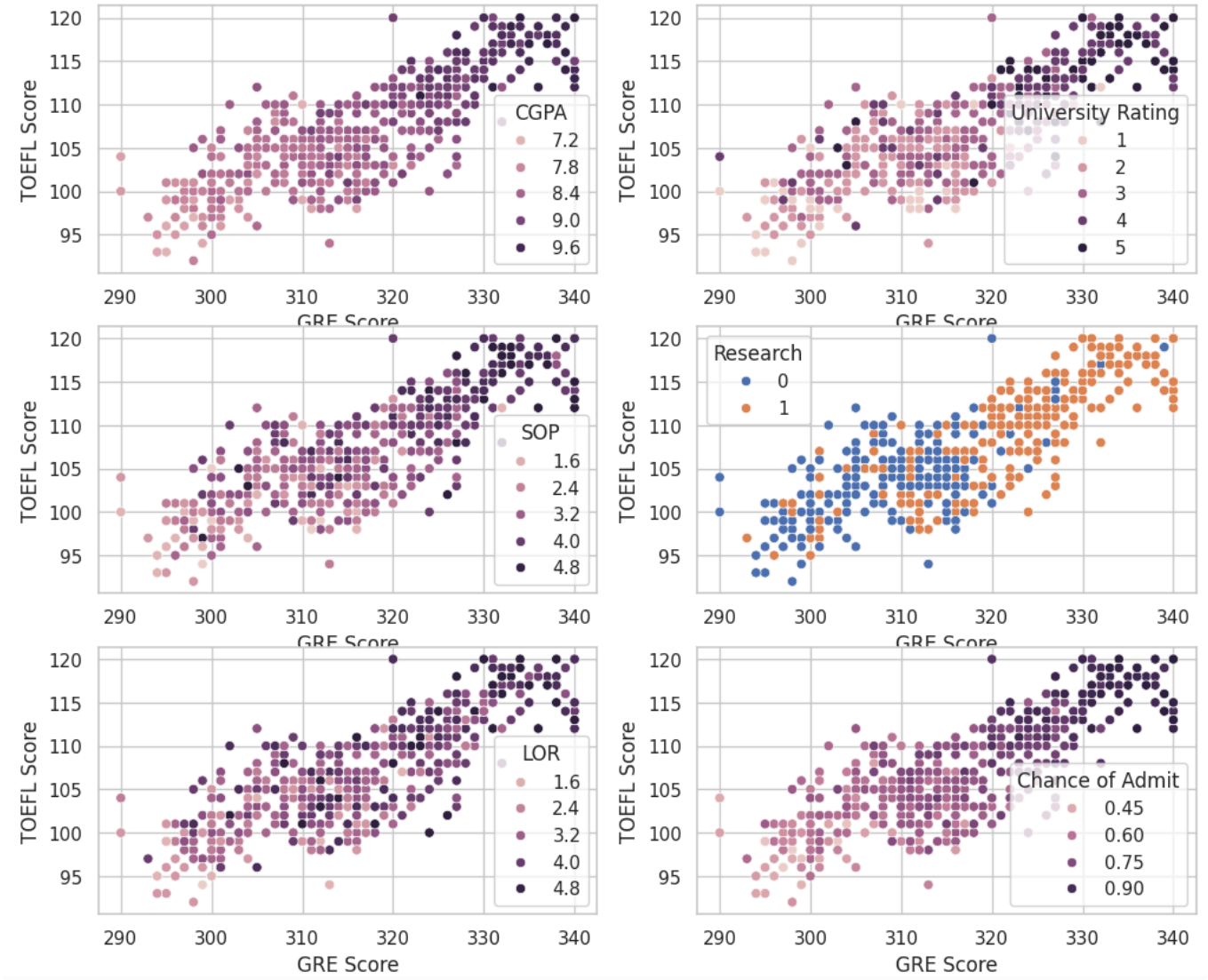
6.cgpa-8 to 9 had perfect uniform dist.

7.>250 students having research experience

Bivariate Analysis

```
fig, axs = plt.subplots(nrows=3, ncols=2, figsize=(12,10))
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="CGPA",ax=axs[0,0])
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="University Rating",ax=axs[0,1])
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="SOP",ax=axs[1,0])
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="Research",ax=axs[1,1])
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="LOR",ax=axs[2,0]) # Removed the extra space after 'LOR'
sns.scatterplot(data=df, x="GRE Score", y="TOEFL Score", hue="Chance of Admit",ax=axs[2,1]) # Removed the extra space after 'Admit'
```

```
<Axes: xlabel='GRE Score', ylabel='TOEFL Score'>
```



Observations from scatter plot:

1.Gre and toe scores follows perfect positive correlation.

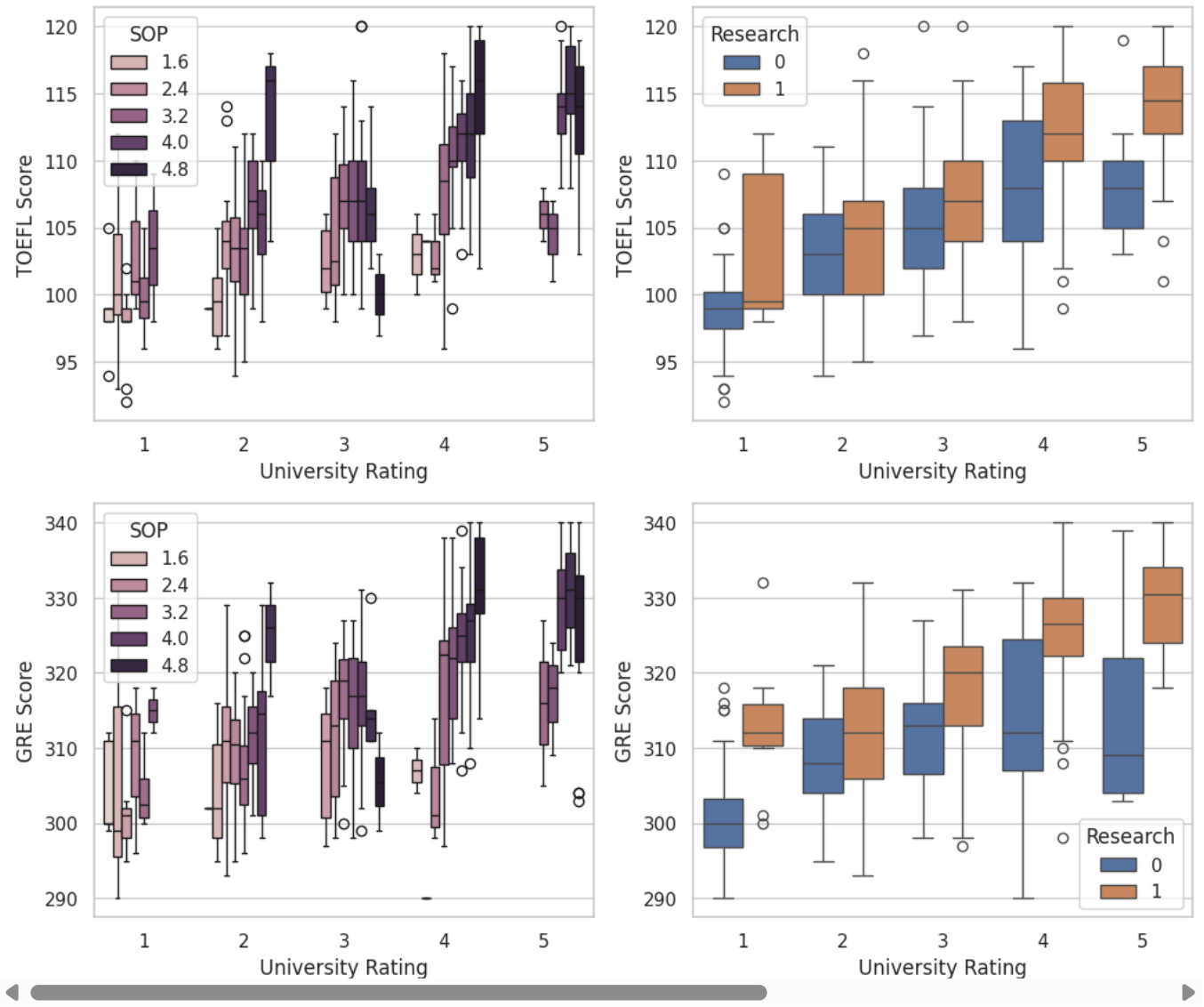
2.For a student both gre and toe scores are in top range.

3.Students having high cgpa,sop,research exposure along with high gre and toe scores are having high chances to admit in top university rating.

4.students who secured toe score>105 and gre score>320 having high chance of admit

```
fig, axs = plt.subplots(nrows=2, ncols=2, figsize=(12,10))
sns.boxplot(x='University Rating', y='TOEFL Score', data=df,hue='SOP',ax=axs[0,0])
sns.boxplot(x='University Rating', y='TOEFL Score', data=df,hue='Research',ax=axs[0,1])
sns.boxplot(x='University Rating', y='GRE Score', data=df,hue='SOP',ax=axs[1,0])
sns.boxplot(x='University Rating', y='GRE Score', data=df,hue='Research',ax=axs[1,1])
```

```
<Axes: xlabel='University Rating', ylabel='GRE Score'>
```



Observations from box plot:

1.students with toe score>110 with sop rating 5, having high chance of getting admission with university ratings 4 and 5.

2.students with toe score>110 having high research exposure

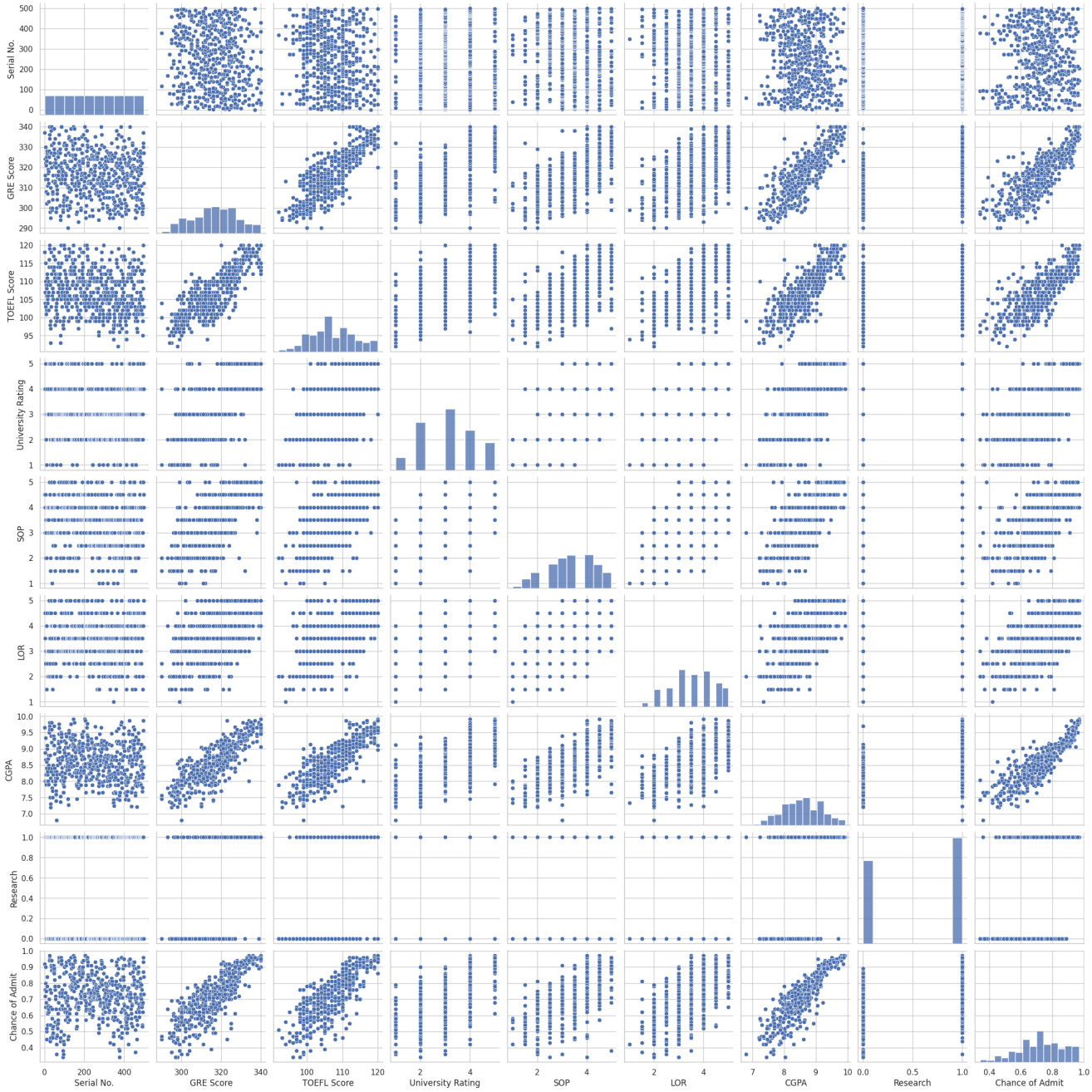
3.students with gre score>320 with sop rating 5, having high chance of getting admission with university ratings 4 and 5.

4.students with gre score>320 having high research exposure

Multivariate Analysis

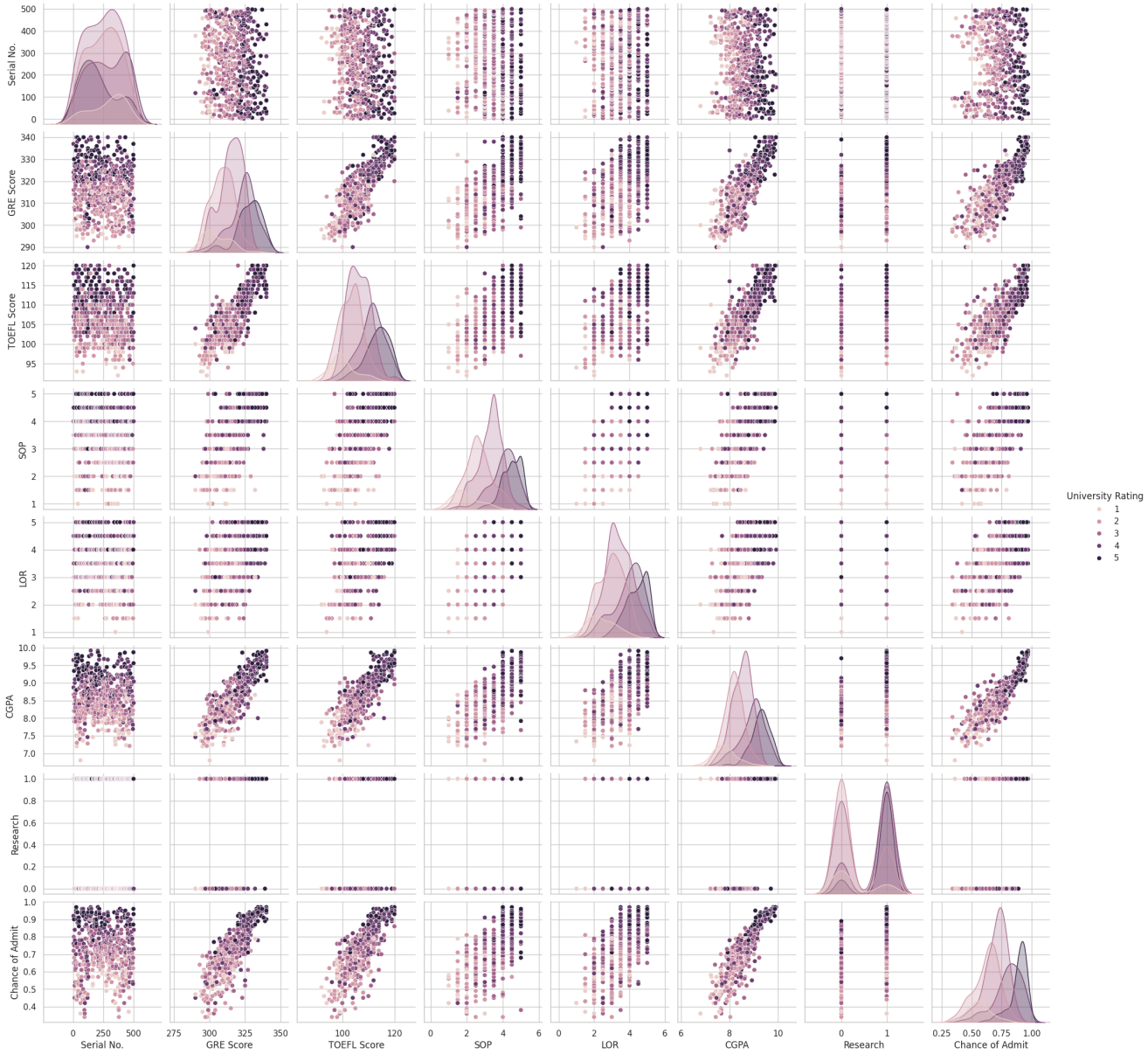
```
sns.pairplot(df)
```

```
<seaborn.axisgrid.PairGrid at 0x7b04702a3c50>
```



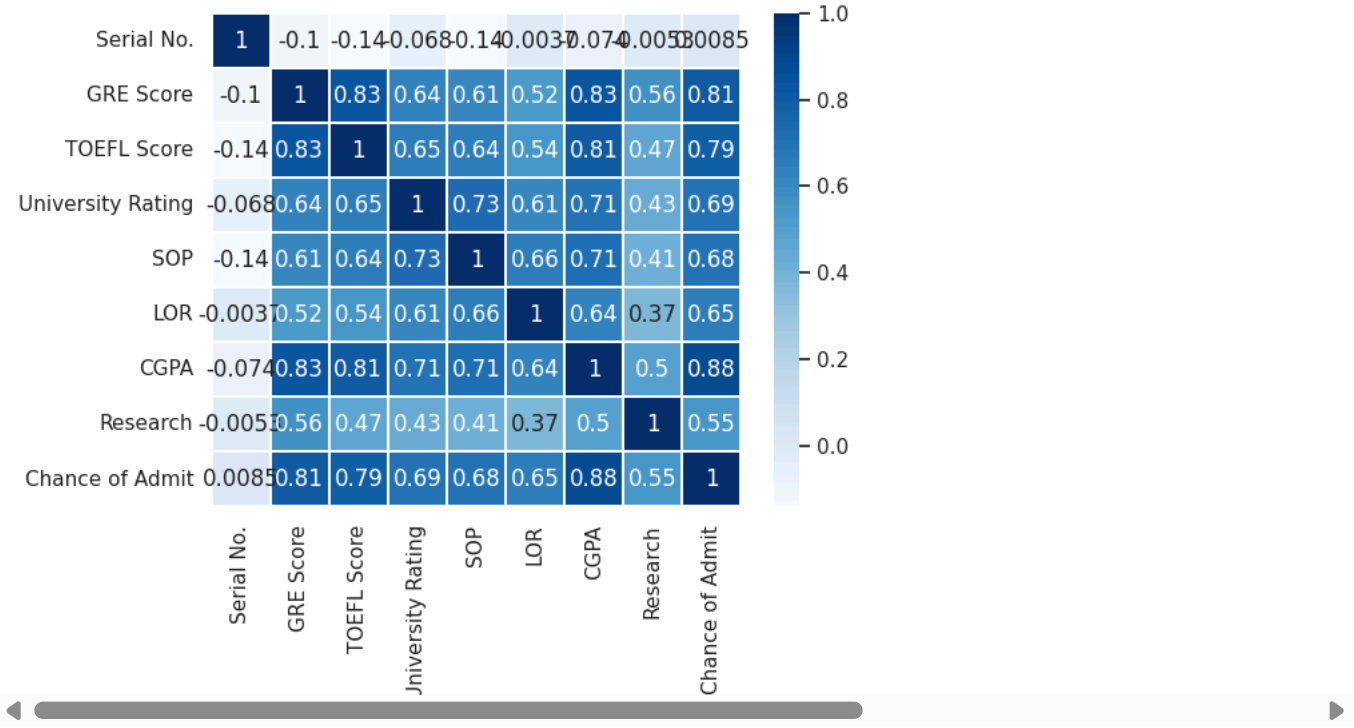

```
sns.pairplot(df,hue="University Rating")
```

```
<seaborn.axisgrid.PairGrid at 0x7b0470523c10>
```




```
sns.heatmap(df.corr(), annot=True, cmap="Blues",linewidth=.1 )
```

<Axes: >



observations:

1.Toe and gre scores having high corr coef = 0.83 we can observe the variation in correlation coe cients wrt different features comparisions.

2.grescore vs chance of admit having high corr coef = 0.81

3.cgpa vs chance of admit having high corr coef = 0.88

2.Data Pre-Processing

```
# Duplicate value check
duplicate_rows = df[df.duplicated()]
print("Number of duplicate rows:", duplicate_rows.shape[0])
# Drop duplicate rows
df = df.drop_duplicates()

# Check for missing values
missing_values = df.isnull().sum()
print("Missing Values:\n", missing_values)
```

```
Number of duplicate rows: 0
Missing Values:
 Serial No.           0
GRE Score             0
TOEFL Score           0
University Rating     0
SOP                   0
LOR                   0
CGPA                  0
Research              0
Chance of Admit       0
dtype: int64
```

There are no missing values in df

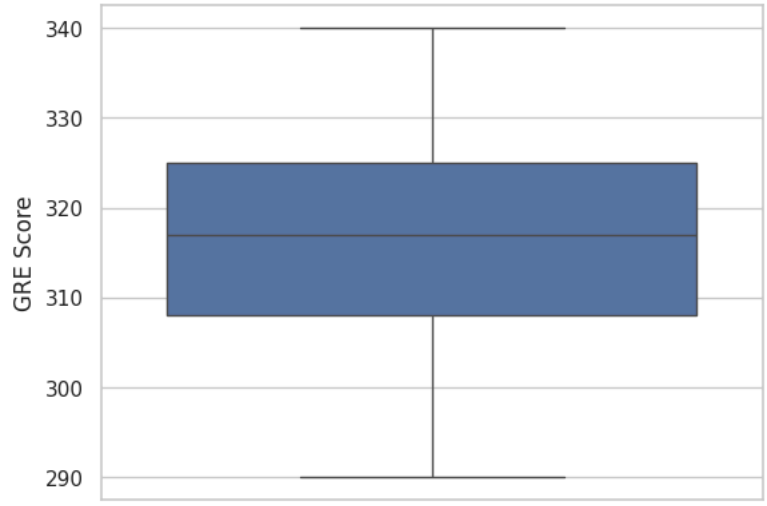
```
# Outlier detection by IQR method
q1= df['GRE Score'].quantile(0.25)
q3= df['GRE Score'].quantile(0.75)
iqr = q3-q1 # iqr- inter quatile range
print('q1','q3','iqr:',q1,q3,iqr)
lower_limit = q1 - (1.5*iqr)
upper_limit = q3 + (1.5*iqr)
print('lower_limit' ',' 'upper_limit:', lower_limit, upper_limit)
#trimming - delete outlier data
new_df = df.loc[(df['GRE Score'] < upper_limit) & (df['GRE Score'] > lower_limit)]
print('before removing outliers:', len(df))
print('after removing outliers:', len(new_df))
print('num of outliers:',len(df)-len(new_df))
```

```
#capping - upper,lower limit
new_df.loc[(new_df['GRE Score'] > upper_limit), 'GRE Score'] = upper_limit
new_df.loc[(new_df['GRE Score'] < lower_limit), 'GRE Score'] = lower_limit
print(new_df)
sns.boxplot(new_df['GRE Score'])
```

```
q1 q3 iqr: 308.0 325.0 17.0
lower_limit,upper_limit: 282.5 350.5
before removing outliers: 500
after removing outliers: 500
num of outliers: 0
     Serial No.  GRE Score  TOEFL Score  University Rating  SOP  LOR  CGPA  \
0             1      337.0          118                  4  4.5  4.5  9.65
1             2      324.0          107                  4  4.0  4.5  8.87
2             3      316.0          104                  3  3.0  3.5  8.00
3             4      322.0          110                  3  3.5  2.5  8.67
4             5      314.0          103                  2  2.0  3.0  8.21
..          ...        ...          ...                ...  ...  ...   ...
495         496      332.0          108                  5  4.5  4.0  9.02
496         497      337.0          117                  5  5.0  5.0  9.87
497         498      330.0          120                  5  4.5  5.0  9.56
498         499      312.0          103                  4  4.0  5.0  8.43
499         500      327.0          113                  4  4.5  4.5  9.04

     Research  Chance of Admit
0           1             0.92
1           1             0.76
2           1             0.72
3           1             0.80
4           0             0.65
..        ...              ...
495         1             0.87
496         1             0.96
497         1             0.93
498         0             0.73
499         0             0.84

[500 rows x 9 columns]
<ipython-input-34-1209a9cd4d93>:15: FutureWarning: Setting an item of incompatible dtype is deprecated and will raise an error in
  new_df.loc[(new_df['GRE Score'] > upper_limit), 'GRE Score'] = upper_limit
<Axes: ylabel='GRE Score'>
```



As there are no outliers in the data ,after performing IQR outlier detection method we can see no of outliers is 0

```
# Feature engineering - Creating a new feature (e.g., Total Scores)
df['Total Scores'] = df['GRE Score'] + df['TOEFL Score']
df.head()
# Dropping unnecessary columns
df = df.drop(['Serial No.'], axis=1, errors='ignore')
df
```

	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit	Total Scores
0	337	118	4	4.5	4.5	9.65	1	0.92	455
1	324	107	4	4.0	4.5	8.87	1	0.76	431
2	316	104	3	3.0	3.5	8.00	1	0.72	420
3	322	110	3	3.5	2.5	8.67	1	0.80	432
4	314	103	2	2.0	3.0	8.21	0	0.65	417
...	...	...	...	...	...	...	...	...	...
495	332	108	5	4.5	4.0	9.02	1	0.87	440
496	337	117	5	5.0	5.0	9.87	1	0.96	454
497	330	120	5	4.5	5.0	9.56	1	0.93	450
498	312	103	4	4.0	5.0	8.43	0	0.73	415
499	327	113	4	4.5	4.5	9.04	0	0.84	440

500 rows × 9 columns

```
df = df.to_csv("result.csv")
```

```
df = pd.read_csv("result.csv")
df.head()
```

	Unnamed: 0	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit	Total Scores
0	0	337	118	4	4.5	4.5	9.65	1	0.92	455
1	1	324	107	4	4.0	4.5	8.87	1	0.76	431
2	2	316	104	3	3.0	3.5	8.00	1	0.72	420
3	3	322	110	3	3.5	2.5	8.67	1	0.80	432
4	4	314	103	2	2.0	3.0	8.21	0	0.65	417

3.Model Building

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error
from statsmodels.stats.outliers_influence import variance_inflation_factor
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from scipy import stats
# Data Preparation for Modeling:
# Separate independent variables (X) and the dependent variable (y)
X = df.drop('Chance of Admit', axis=1)
y = df['Chance of Admit']
X.shape, y.shape
```

```
((500, 9), (500,))
```

```
# Split the dataset into training and testing sets
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print('X_train:', X_train.shape)
print('y_train:',y_train.shape)
print('X_test:',X_test.shape)
print('y_test:',y_test.shape)
```

```
X_train: (400, 9)
y_train: (400,)
X_test: (100, 9)
y_test: (100,)
```

```
y_train = y_train.values.reshape(400,-1)
y_test = y_test.values.reshape(100,-1)
```

We will use

X_train and y_train to train our model.

X_test and y_test to evaluate the model's performance

Scaling the data

```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df = pd.DataFrame(scaler.fit_transform(df), columns=df.columns)
df.head()
```

	Unnamed: 0	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit	Total Scores
0	0.000000	0.94	0.928571	0.75	0.875	0.875	0.913462	1.0	0.920635	0.931507
1	0.002004	0.68	0.535714	0.75	0.750	0.875	0.663462	1.0	0.666667	0.602740
2	0.004008	0.52	0.428571	0.50	0.500	0.625	0.384615	1.0	0.603175	0.452055
3	0.006012	0.64	0.642857	0.50	0.625	0.375	0.599359	1.0	0.730159	0.616438
4	0.008016	0.48	0.392857	0.25	0.250	0.500	0.451923	0.0	0.492063	0.410959

```
# Build and train the Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
LinearRegression()
```

```
# Make predictions on the test set
y_pred = model.predict(X_test)
```

```
# Model statistics
from sklearn.metrics import mean_squared_error, r2_score
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print(f'Mean Squared Error: {mse}')
print(f'R-squared: {r2}')
```

```
Mean Squared Error: 0.003551180545098705
R-squared: 0.8263481396039754
```

```
ridge_model = Ridge(alpha=1.0)
ridge_model.fit(X_train, y_train)
lasso_model = Lasso(alpha=1.0)
lasso_model.fit(X_train, y_train)
```

```
Lasso()
```

4.Testing the assumptions of the linear regression model

```
#4.1.Multicollinearity check by VIF score
X_sm = sm.add_constant(X_train)  # Statmodels default is without intercept, to add intercept we need to add constant.
model = sm.OLS(y_train, X_sm)
results = model.fit()
# Print the summary statistics of the model
print(results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.829
Model:                            OLS   Adj. R-squared:                  0.825
Method:                 Least Squares   F-statistic:                     236.4
Date:                Sat, 15 Mar 2025   Prob (F-statistic):          1.20e-144
Time:                        11:12:49   Log-Likelihood:                 570.60
No. Observations:                 400   AIC:                            -1123.
Df Residuals:                     391   BIC:                            -1087.
Df Model:                           8
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0
------------------------------------------------------------------------------
const          -1.4754      0.121    -12.165      0.000      -1.714      -1.
Unnamed: 0   8.938e-05   2.15e-05      4.166      0.000    4.72e-05       0.000
```

```
GRE Score            0.0005      0.001      0.870      0.385      -0.001       0.002
TOEFL Score          0.0015      0.001      2.022      0.044    4.05e-05       0.003
University Rating    0.0026      0.004      0.639      0.523      -0.005       0.011
SOP                  0.0042      0.005      0.845      0.399      -0.006       0.014
LOR                  0.0150      0.005      3.306      0.001       0.006       0.024
CGPA                 0.1091      0.011     10.300      0.000       0.088       0.130
Research             0.0223      0.007      3.054      0.002       0.008       0.037
Total Scores         0.0020      0.000      6.939      0.000       0.001       0.003
==============================================================================
Omnibus:                       63.061   Durbin-Watson:                   2.062
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              112.681
Skew:                          -0.907   Prob(JB):                     3.40e-25
Kurtosis:                       4.863   Cond. No.                     3.43e+15
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.22e-23. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_tr_scaled = scaler.fit_transform(X_train)
X_sm = sm.add_constant(X_tr_scaled)  #Statmodels default is without intercept, to add intercept we need to add constant
sm_model = sm.OLS(y_train, X_sm).fit()
print(sm_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.829
Model:                            OLS   Adj. R-squared:                  0.825
Method:                 Least Squares   F-statistic:                     236.4
Date:                Sat, 15 Mar 2025   Prob (F-statistic):          1.20e-144
Time:                        11:13:06   Log-Likelihood:                 570.60
No. Observations:                 400   AIC:                            -1123.
Df Residuals:                     391   BIC:                            -1087.
Df Model:                           8
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.7242      0.003    246.426      0.000       0.718       0.730
x1             0.0125      0.003      4.166      0.000       0.007       0.018
x2             0.0162      0.005      3.291      0.001       0.007       0.026
x3             0.0148      0.006      2.611      0.009       0.004       0.026
x4             0.0030      0.005      0.639      0.523      -0.006       0.012
x5             0.0042      0.005      0.845      0.399      -0.006       0.014
x6             0.0138      0.004      3.306      0.001       0.006       0.022
x7             0.0655      0.006     10.300      0.000       0.053       0.078
x8             0.0111      0.004      3.054      0.002       0.004       0.018
x9             0.0164      0.002      6.904      0.000       0.012       0.021
==============================================================================
Omnibus:                       63.061   Durbin-Watson:                   2.062
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              112.681
Skew:                          -0.907   Prob(JB):                     3.40e-25
Kurtosis:                       4.863   Cond. No.                     1.76e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 7.21e-30. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

```
vif = pd.DataFrame()
X_t = pd.DataFrame(X_tr_scaled, columns=X_train.columns)
vif['Features'] = X_t.columns
vif['VIF'] = [variance_inflation_factor(X_t.values, i) for i in range(X_t.shape[1])]
vif['VIF'] = round(vif['VIF'], 2)
vif = vif.sort_values(by = "VIF", ascending = False)
vif
```

```
/usr/local/lib/python3.11/dist-packages/statsmodels/stats/outliers_influence.py:197: RuntimeWarning: divide by zero encountered in s
  vif = 1. / (1. - r_squared_i)
```

	Features	VIF
1	GRE Score	inf
2	TOEFL Score	inf
8	Total Scores	inf
6	CGPA	4.68
4	SOP	2.82
3	University Rating	2.57
5	LOR	2.01
7	Research	1.52
0	Unnamed: 0	1.05

There is no multi collinearity in the data as we can observe that VIF<5

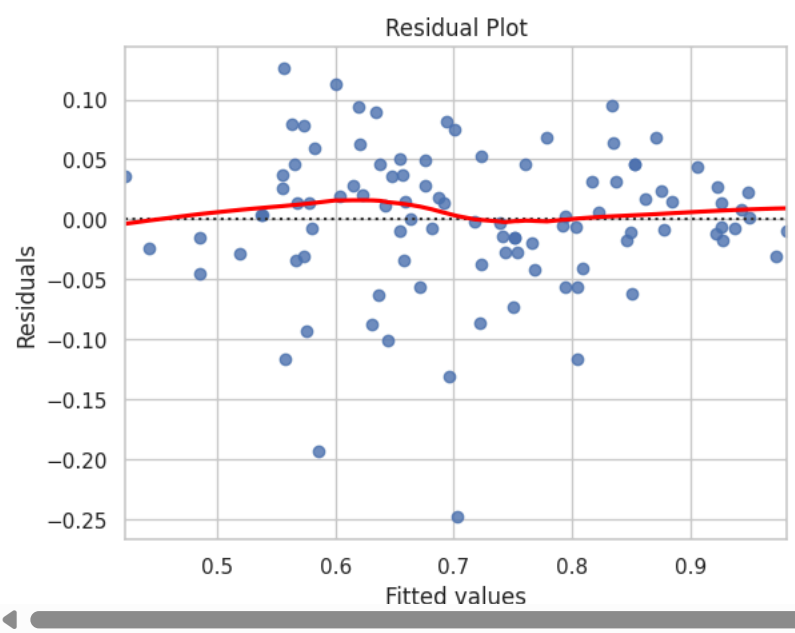
```
#4.2 The mean of residuals is nearly zero
# Add a constant column to X_test for prediction to match the training data
X_test_sm = sm.add_constant(X_test)
# Predict using the modified X_test and the statsmodels model (sm_model or results)
# Choose the model you want to evaluate (sm_model or results)
y_pred = results.predict(X_test_sm)  # If using the model from cell 54
# OR
#y_pred = sm_model.predict(X_test_sm) # If using the model from cell 55

# Reshape y_pred to have the same dimensions as y_test
y_pred = y_pred.values.reshape(y_test.shape)

residuals = y_test - y_pred
mean_residuals = np.mean(residuals)
print(f"Mean of Residuals: {mean_residuals}")
```

```
Mean of Residuals: -0.005711474436268695
```

```
#4.3 Linearity of variables (no pattern in the residual plot)
sns.residplot(x=y_pred, y=residuals, lowess=True, line_kws={'color': 'red'})
plt.xlabel('Fitted values')
plt.ylabel('Residuals')
plt.title('Residual Plot')
plt.show()
```




```
# 4.4 Test for Homoscedasticity
 # Plot residuals against predicted values
plt.scatter(y_pred, residuals)
plt.title('Residual Plot')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.show()
```